STUDY ON DEVELOPMENT, SENSORY EVALUATION AND STORAGE ABILITY ASSESMENT OF CHOCOLATE FLAVORED MILK

A Project report

By

KHAIRUZZAMAN BABU	**MD. AMAN ULLAH**	**NASIM AL MASUD**
St. ID: 18-2-04-048	St. ID: 18-2-04-047	St. ID: 18-2-04-25
Session: 2018-2019	Session: 2018-2019	Session: 2018-2019

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Department of Food Science & Engineering

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<u>Submitted To</u>

The department of Food Science and Engineering

German University Bangladesh Gazipur-1702

In partial fulfillment of the requirements for the degree of B.Sc.

In Food Science and Engineering

January 2022

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DECLARATION

We thereby declare that the project work entitled "Study on 'Study and development of chocolate flavored milk. Submitted to the department of Food Science& Engineering, German University Bangladesh. This project work is submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Food Science and Engineering carried out by us under the supervision of **Bejoya Das Tumpa**, Lecturer, Department of Food Science and Engineering, German University Bangladesh. The result embodied in this thesis which has not been submitted to any other institute.

January- 2022
THE AUTHOR

IN THE NAME OF ALMIGHTY GOD, THE MOST GRACIOUS, THE MOST MERCIFUL

Dedicated to our beloved parents & all well-wishers,
Thank you for your never-ending inspiration and love

ABSTRACT

The present study was conducted to develop shelf stable flavored milk. In order to determine the most stable flavored milk product for ($S_{1=}$9.0% sugar), (S_2=8.0% sugar), and (S_3=7.3% sugar), samples were stored at freezing temperature using poly pack and tetra pack for 07 days. Sensory evaluation of flavored milk revealed that sample S_1 had better acceptability than those of sample S_2 and sample S_3. The storage studies revealed that the flavored milk stored in tetra pack had better quality in terms of color, flavor and texture. The various constituent of flavored milk such as acidity, °Brix, fat%, p^H and total solid were determined. During storage after 07 days the better shelf life of flavored milk was found in sample S_1 which was packed in tetra packaging system.

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CHAPTER-I

1.1 Introduction

Nutrient-rich flavored milk comes in a variety of flavors and offers the same unique nutrient package of nine essential nutrients as unflavored milk. Flavored milk is available in traditional flavors such as chocolate as well as innovative flavors including strawberry, vanilla, mocha and root beer. As the most popular milk choice in schools, flavored milk is a highly palatable, nourishing beverage that can help Americans, particularly children, meet current daily dairy food and calcium intake recommendations. Despite the important nutrient contributions flavored milk makes to the diet, concerns about the potential effects of the added sugar and flavorings in flavored milk have raised questions regarding the role of flavored milk in a healthy diet. This review highlights flavored milk's valuable contributions to nutrient intake and health, presents recommendations offered by health professional organizations and nutrition experts regarding flavored milk consumption, addresses questions about the consumption of this nutrient-rich beverage and provides data that demonstrates the importance of flavored milk in the school environment.(Teri Gacek Associates (TGA): Flavored Milk & Sugar — Is It An Issue For Moms, Dietitians and Pediatricians? A Qualitative Exploration for Dairy Management Inc. (DMI), November 2005)

1.2 Classification of flavored milk

The number of commercially flavored milks such as chocolate, vanilla, banana, orange and strawberry. Chocolate milk is the most popular flavored milk, especially with children (90% like its taste); it has high protein content and is a good source of vitamins, calcium and other nutrients. Strawberry flavored milk also demon-striates high acceptability with the consumer (Miller, Jarvis, & McBean, 2007). (Miller, G.D., Jarvis, J.K., & McLean, L.D. (2007). Handbook of Foods and Nutrition.)

1.3 Classification of chocolate

Different forms and flavors of chocolate are produced by varying the quantities of the different ingredients. Other flavors can be obtained by varying the time and temperature when roasting the beans.

- "Unsweetened chocolate", also known as "bitter", "baking chocolate" or "cooking chocolate" is pure chocolate liquor mixed with some form of fat to produce a solid substance. The pure, ground, roasted cocoa beans impart a strong, deep chocolate flavor. With the addition of sugar, however, it is used as the base for cakes, brownies, confections, and cookies.
- Dark chocolate", also called "black chocolate", is produced by adding fat and sugar to cocoa. It is chocolate with no milk or much less than milk chocolate. The U.S. has no official definition for dark chocolate but European rules specify a minimum of 35% cocoa solids. Dark chocolate can be eaten as is, or used in cooking, for which thicker, more expensive baking bars with higher cocoa percentages ranging from 70% to 99% are sold. Dark is synonymous with semisweet, and extra dark with bittersweet, although the ratio of cocoa butter to solids may vary.

- "Semisweet chocolate" is frequently used for cooking purposes. It is a dark chocolate with (by definition in Swiss usage) half as much sugar as cocoa, beyond which it is "sweet chocolate."
- Bittersweet chocolate" is chocolate liquor (or unsweetened chocolate) to which some sugar (less than a third), more cocoa butter, vanilla and sometimes lecithin has been added. It has less sugar and more liquor than semisweet chocolate, but the two are interchangeable when baking. Bittersweet and semisweet chocolates are sometimes referred to as 'couverture'. Many brands now

print on the package the percentage of cocoa in the chocolate (as chocolate liquor and added cocoa butter). The higher the percentage of cocoa, the less sweet the chocolate is.

- "Couverture" is a term used for chocolates rich in cocoa butter. Popular brands of couverture used by professional pastry chefs and often sold in gourmet and specialty food stores include: Valrhona, Felchlin, Lindt & Sprüngi, Scharffen Berger, Cacao Barry, Callebaut, Chocolate, Choco fig fuel chocolates and Guittard. These chocolates contain a high percentage of cocoa.

- Milk chocolate is a classic that we all know and love from childhood. With its light brown color, creamy texture, and sweet flavor, milk chocolate is widely regarded as the most popular type of chocolate. It is made by combining chocolate liquor (cocoa solids and cocoa butter) with sugar, and milk. Sometimes an emulsifier, such as soy lecithin, is added to enhance its smoothness. According to the FDA definition, milk chocolate must contain at least 10% chocolate liquor and 12% milk.

- "White chocolate" is a confection based on sugar, milk, and cocoa butter without the cocoa solids.

- "Cocoa powder" is used for baking, and for drinking with added milk and sugar. There are two types of unsweetened cocoa powder: natural cocoa (like the sort produced by the Brome process), and Dutch-process cocoa
- "Compound chocolate" is the technical term for a confection combining cocoa with vegetable fat, usually tropical fats and/or hydrogenated fats, as a replacement for cocoa butter. It is often used for candy bar coatings. In many countries it may not legally be called "chocolate".
- "Raw chocolate" is chocolate that has not been processed, heated, or mixed with other ingredients. It is sold in chocolate-growing countries, and to a much lesser extent in other countries, often promoted as healthy

1.4 General objectives

The main objective of this project is to design, construct and taste of chocolate flavored milk.

1.5 Specific objectives

i) To find out suitable formulations of chocolate flavored milk.

ii) Chemical analysis of chocolate flavored milk.

iii) To study the shelf life of chocolate flavored milk.

iv) To determine the microbiological & chemical parameters of chocolate flavored milk

Chapter II
REVIEW OF LITERATURE

2.1 Dairy Products

Milk and other dairy foods are the major source of calcium in human diet. A glass of milk contains approximately 240 mg of calcium. Milk also provides other essential nutrients such as riboflavin, phosphorus, protein, potassium, vitamin A, vitamin B 12 and niacin. The calcium from dairy products is readily bio-available. An individual's body absorbs about 35% of the calcium contained in a dairy product (Heaney an others, 1988). Calcium digestion is facilitated by lactose ingestion (Schaafsma, 1980). On the other hand the calcium from non-dairy products may be absorbed at only 75% the efficiency of calcium from dairy foods (Heaney and others, 2000). Even though calcium is so important for human health, it is one of the nutrients most likely to be consumed in low amounts by children, adolescents, and adult. Data obtained from the 1994-96 Continuing Survey of Food Intakes by Individuals revealed that 71% of girls between the age of 6 to 11 years and 62% of similar aged boys did not meet 100% of the recommended daily intake of calcium. Among females and males with the age of 12 to 19 years, 88% and 68%, respectively, did not meet 100% of the calcium daily intake recommendation of 1,300mg/day. Adult females failed to meet the recommended dietary allowance of calcium.

Among the young children, consumption of fluid milk decreased by 16% since the late 70's (USDA, 1997). One of the reasons for this decrease in calcium intake may be due to the competition of beverages, particularly soft drinks, fruit drinks and flavored teas (Crane and others, 2000). Children and adolescents have increased their intake of soft drinks, fruit drinks and fruit-flavored drinks at the expense of milk. Lactose into learner is another important reason of why US consumers decline to drink milk.

Nevertheless, new value-added milk products are entering the market, and perhaps replacing the diminished market position of traditional fluid milk. Many of them stand out as being indicators of where the value- added milk products category is headed these years. According to the reports in the Dairy Foods Magazine, in late 1999, Suiza Foods (Dallas, TX) developed value-added

milks geared to very specific c needs, such as its “Milk Made Better” line is fortified with calcium. The 3 varieties include Kids milk (with 50% more vitamins), Fit milk (which is fat free but has the taste of low fat milk), and Life milk (for the 40+ age group). Dannon (Tarrytown, NY) brought its “Actimel” dairy-based dietary supplement from Europe to the U.S. It contains "a combination of beneficial cultures" and should be taken every morning to "keep the body at its best." Lancashire Dairies introduced in April, 2002 flavored milks with banana, chocolate and strawberry flavors.

The Havana Cappuccino Co has just launched to the market the new ready-to-drink Royal Mandalay Chai™, which is made with evaporated milk and a blend of black, green and chamomile teas with spices of cinnamon, ginger and clove. Cholesterol-reducing margarine spreads also made a big splash in the dairy category. Unilever Foods (Greenwich, CT) and McNeil Specialty Products (Fort Washington, PA) introduced margarines made with plant stanol esters, which help reduce blood serum cholesterol. Unilever’s Take Control and McNeil’s Benecol started with just margarine spreads (at three times the price of regular margarine) and branched out to salad dressings and snack bars.

Perhaps the most innovative among new products in the dairy industry today is the New Bottle Can (a re-closable aluminum beverage container that ranges in size from 300-500 ml). It is manufactured by Japan-based Daiwa Can Co., and Mitsui & Co., Ltd., and will initially be imported to the United States. Ball Corp. (Broomfield, Colorado) has the exclusive rights to sell and distribute the container in the US (Berry, 2003)

Cheese introductions include the usual lineup of healthy, imported, pre-cut, shredded, and flavored varieties. Two companies offered cheese in unusual forms. Crystal Farms , (Minneapolis) offered cheese in an aerosol can under the Cheezoids label, and Ariza Cheese , (Paramount, California) introduced Cheese U Squeeze, which comes in a toothpaste-like tube (Dairy Foods, 2002, 2003)

The 2002 consumption of whole milk powder in the US was 26 million tons compared with that in 1998 (52 million tons). On the other hand, non-fat milk powder had an increase in its consumption during 2002 of about 20%, compared with that of 1998 (USDA, 2002).

Milk powder is produced by removing water from pasteurized, homogenized liquid milk (whole fat, partially skimmed, or fat free), either by roller drying, freeze-drying, or spray-drying processes (Early, 1998). It possesses all the appealing qualities of milk and, in its dry form, is an important ingredient for manufacturing a wide range of food products. Whole milk powder has a shelf life of about 6 months (Early, 1998). Milk powder is a milk ingredient considered to be natural, nutritious, cost-effective and beneficial in its functionality. It has a particularly wide range of applications in the confectionery industry, namely for chocolate, toffee and caramel confections, ice cream, milk beverages, toppings and icings, recipe dishes, infant formula, breakfast cereals, soups and sauces, bakery products and coffee whiteners (Early, 1998).

Milk powder has been used mainly as a food ingredient, but there is still a gap, which leads to the opportunity to arrive with a new food product of which milk powder could constitute the major proportion.

2.2 History of Chocolate Flavored Milk

Chocolate milk is a sweetened, usually cold, cocoa-flavored milk drink. It is created when chocolate syrup (or chocolate powder) is mixed with milk (from cows, goats, soy, rice, etc.). It can be purchased pre-mixed or made at home with both cocoa powder and a sweetener (such as sugar or a sugar substitute), or with melted chocolate, chocolate syrup, or chocolate milk mix. Other ingredients, such as starch, salt, carrageenan, vanilla, or artificial flavoring may be added. The carrageenan is used at very low concentrations to form an imperceptible weak gel that prevents the large, dense particles of chocolate from sedimenting. Chocolate milk should be refrigerated like plain milk. It was invented by Hans Sloane in the late 1680s.

Chocolate milk

Some nutritionists have criticized chocolate milk for its high sugar content and its relationship to childhood obesity. Of the milk served in U.S. schools, 71 percent is flavored. In New York City, school food officials say nearly 60 percent of the 100 million cartons served each year are fat-free chocolate milk. Because chocolate milk can contain twice as much sugar as plain low-fat milk from added sugars, some school districts have stopped serving it, including those in Berkeley, CA, and Washington, DC. In the US, 32 percent of children and teens are overweight and at risk for diabetes, high blood pressure and other issues related to obesity.

A number of studies indicate that chocolate milk aids in recovery when taken after athletic workouts. A study at Loughborough University found that milk was an effective rehydration drink. In a 2010 report it was studied with 13 male college soccer players with results in favor of chocolate milk workout supplementation. A study in 2011 at Kean University in New Jersey concluded similar results in male soccer players discovering there was even an increased time to fatigue when chocolate milk was consumed. The Kean University study also viewed chocolate milk's effects on women soccer players undergoing morning and afternoon practices during preseason. They were given either the carbohydrate-electrolyte beverage or chocolate milk between morning and afternoon preseason practices. Following every afternoon practice, each athlete completed a shuttle run to fatigue. The study concluded that chocolate milk is just as

beneficial as the carbohydrate-electrolyte beverage in promoting recovery in women. In a 2006 study, authors believed the benefits were due to its ratio of carbohydrates to protein, among other nutritional properties. However, this study was small in scale with only nine athletes and partially funded by the dairy industry. Furthermore, the study compared chocolate milk to two energy drinks and unflavored milk was not used as a comparison, so it is unknown if chocolate milk is superior to unflavored milk as a recovery drink.

A 2005 study by the New York City (NYC) Department of Education found that by removing whole milk and replacing it with low-fat to fat-free chocolate milk, students were served an estimated 5,960 fewer calories and 619 fewer grams of fat per year.

Chocolate supplies oxalic acid, which reacts with the calcium in the milk producing calcium oxalate, thus preventing the calcium from being absorbed in the intestine. However, it is present in small enough amounts that the effect on calcium absorption is negligible. As chocolate contains relatively small amounts of oxalate, it is unclear to what extent chocolate consumption affects healthy people who eat calcium-rich diets. In a 2008 study, participants who consumed one or more servings of chocolate on a daily basis had lower bone density and strength than those participants who ate a serving of chocolate six times a week or less. Researchers believe this may be due to oxalate inhibiting calcium absorption — but it could also be due to the chocolate's sugar content, which may increase calcium excretion. It is clear, however, that consuming foods high in oxalate — and in turn their effect on calcium absorption — is a more significant concern for people with oxalate kidney stones, which occur when there is too much oxalate in the urine. These people, especially, should reduce their oxalate intake and increase their calcium intake. However, the high magnesium content in chocolate is likely to reduce the risk of stone formation somewhat, because like citrate, magnesium is also an inhibitor of urinary crystal formation.

A November 2009 study conducted by scientists in Barcelona, Spain suggests that regularly consuming skim milk with cocoa rich in flavonoids may reduce inflammation and slow or prevent the development of atherosclerosis. However, the study notes that its effects are not as pronounced as seen in consumption of red wine.

The tasty secret of the cacao (kah KOW) tree was discovered 2,000 years ago in the tropical rainforests of the Americas. The pods of this tree contain seeds that can be processed into chocolate. The story of how chocolate grew from a local Mesoamerican beverage into a global sweet encompasses many cultures and continents.

The first people known to have made chocolate were the ancient cultures of Mexico and Central America. These people, including the Maya and Aztec, mixed ground cacao seeds with various seasonings to make a spicy, frothy drink.

Later, the Spanish conquistadors brought the seeds back home to Spain, where new recipes were created. Eventually, and the drink's popularity spread throughout Europe. Since then, new technologies and innovations have changed the texture and taste of chocolate, but it still remains one of the world's favorite flavors.

CHAPTER-III

MATERIALS & METHODS

3.1 Raw Materials

1. Milk
2. Sugar
3. Chocolate
4. Powder
5. Concentrate
6. SMP
7. Stabilizer
8. Water
9. Color
10. Flavor.

3.2 Specification of raw materials

- **Milk**

Milk is a white liquid produced by the mammary glands of mammals. It is the primary source of nutrition for young mammals before they are able to digest other types of food. Early-lactation milk contains colostrums, which carries the mother's antibodies to the baby and can reduce the risk of many diseases in the baby. It also contains many other nutrients.

In general, the gross composition of cow's milk in the U.S. is 87.7% water, 4.9% lactose (carbohydrate), 3.4% fat, 3.3% protein, and 0.7% minerals (referred to as ash). Milk composition varies depending on the species (cow, goat, sheep), breed (Holstein, Jersey), the animal's feed, and the stage of lactation. Although there are minor variations in milk composition, the milk from different cows is stored together in bulk tanks and provides a relatively consistent composition of milk year round in the U.S (U.S. Dept. of Health and Human Services, Public Health Service and Food and Drug Admin. Grade “A” Pasteurized Milk Ordinance. 2015 Revision).

- **Sugar**

Sugar is the generalized name for a class of chemically-related sweet-flavored substances, most of which are used as food. They are composed of carbon, hydrogen and oxygen. There are various types of sugar derived from different sources.

- **Chocolate Flavored**

For those that like to indulge in their sweet tooth and chocolate cravings, Coffee AM is proud to present a diverse collection of chocolate-inspired coffees. Whether you love a plain chocolate flavor, such as the Swiss Chocolate Flavored coffee, or you want something fruity, such as the Chocolate Strawberry Flavored Coffee – we have something that will make your mouth water!

Made with 100% authentic Arabica beans, these chocolate coffee creations will be of the highest quality with optimal freshness and flavor. Tooth first thing in the morning or after-dinner, Coffee AM certainly will hit the spot with its incredible choices that will tempt your taste buds and satisfy your chocolate desires!

Chocolate Flavor one of all time favorites. I have used this chocolate flavor to make many things. From drinks to sauces this chocolate flavor is one of our most versatile flavors. This chocolate flavor is great for baking and does extremely well when baked in cakes and cookies. I have used this chocolate flavor to make many variations of ice cream and sorbets. Remember like all flavors chocolate flavor needs to drive the flavor otherwise it will taste to candy like. I have used this chocolate flavor to make many chocolate flavored sauces and I have enhanced chocolate sauces that contain a large amount of cocoa so that it will have that just picked taste for long term freezing and storage.

3.3 Packing Materials

- **List of packing materials**

1. Poly pack,
2. Tetra pack etc.

- **Specification of packing materials**

Packaging Material for Poly film

There is 3-layer, 5-layer co-extruded poly packaging film available for customers. The packaging film is suitable for pouches sealed by 3 sides and pouches with middle sealing at the back of the pouches.

- **Application**

  1) High barrier packaging film for the plastic pouch aseptic packaging of UHT milk, milk-based beverage, and other liquid food.
  2) High temperature steaming & boiling packaging film for the plastic pouch packaging of milk and other liquid food.
  3) Packaging film for the plastic pouch packaging of pasteurized milk, yogurt.
  4) High temperature steaming & boiling capping film.

Milk Packaging Film: Pasteurized Milk Pouch, Yogurt Pouch, UHT.

- **Aseptic Tetra Pack Packaging**

The aseptic packaging implies that both packaging and filling environment be sterilized thus preventing any recontamination of the food after being "purified" through thermal treatment. References (Trevor J. Britz University of Stellenbosch South Africa Richard K. Robinson Consultant in Food Science and Technology Reading UK) Furthermore, if we talk about UHT milk, we must know that a multilayer cardboard packaging is used, which also contains an aluminum foil in the middle, captured between the inner layers of cardboard and polyethylene. The cardboard box protects the milk from contact with light and oxygen, two factors that contribute to vitamin degradation. Vitamin B_2 in milk exposed to light for 12 hours for example in a transparent recipient, degrades up to 35%.

Aseptic technology is based on the invention of the Swedish company Tetra Pak, which created the first aseptic packaging in 1961. This packaging allows storing milk or other perishable liquids for a period of 3 - 6 months, without adding preservatives or refrigerating the product. By forming a barrier and impeding oxygen from penetrating (the packaging consists of multiple cardboard, aluminum and polyethylene layers for the food industry) and using a special technique to sterilize the packaging, aseptic technology ensures the protection of food against environmental aggressions. The International Institute for Food Technologies declared the technology of aseptic processing to be "the most important innovation of the 20th century in the industry".

3.4 Table :1 Formulation of mixed chocolate flavored milk

Sample	S1	S2	S3
Milk (%)	79.9%	80.7%	81.4%
Sugar (%)	9.0%	8.0%	7.3%
Milk Gel (%)	0.2%	0.2%	0.2%
SMP (%)	2.5%	2.5%	2.6%
Chocolate concentrate (%)	5.3%	5.4%	5.4%
Chocolate color (%)	0.001%	0.001%	0.001%
Chocolate Flavor (%)	0.04%	0.04%	0.04%
Water (%)	3.0%	3.0%	3.1%
Total (%)	100.0%	100.0%	100.0%

3.5 Chocolate Flavored Milk

Procedures

- Standardize the milk of requirement in a blender (Pasteurization at 80-85°c).
- Add required quantity of SMP and mixing by the blender.
- Add required quantity of sugar, treated chocolate concentrate, color, stabilizer (as premix specification) on the standard milk and mix well.

- Then add flavor with continuous blending for 10-25 min.
- Fill and packs desired.
- Store at milk storage temperature (below 4°c) for dispatching.

Production process of Chocolate Flavored Milk

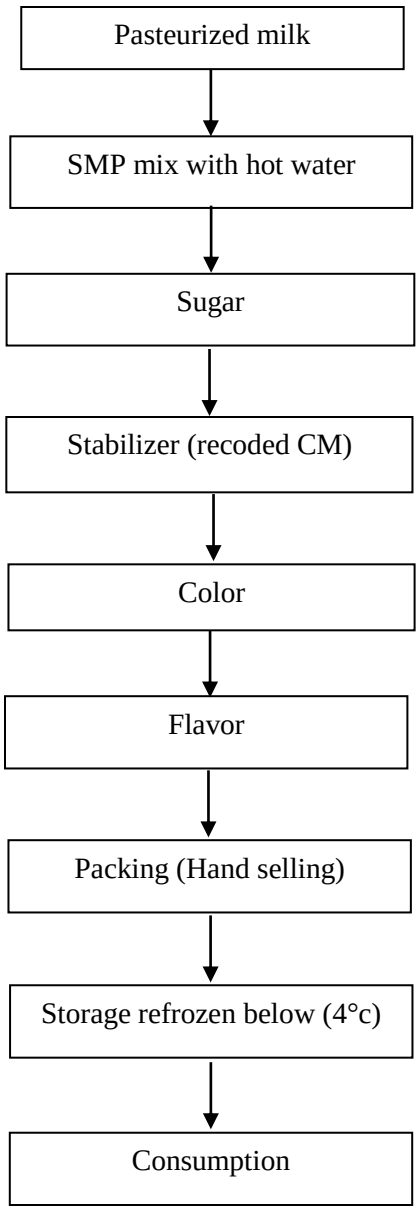


Figure: 1. Production procedure of chocolate flavored milk

Chemical Analysis

3.7 Solid Test

The milk samples were analyzed for total solids contents. For this purpose, fresh milk sample was thoroughly mixed and 2-3g was transferred to a pre-weighed flat bottom dish (AOAC, 1990). After evaporation on steam bath, it was transferred to a hot air oven at 101°C. Dried sample was transferred to a desiccators having silica gel as desiccant. After 1h, the dish was weighed and kept in an oven for further drying. Statistical analysis of data collected was performed using (~30 min). It was again transferred to the desiccators, cooled and weighed as before. The heating, cooling and weighing processes were repeated until constant weight was achieved. Total solids content was calculated by the following formula: the computer programmed SPSS Release 7.5.1 (SPSS, 1996).

$$\textbf{Total solid (\%)} = \frac{\textbf{Weight of dried sample}}{\textbf{Weight of milk sample}} \times 100$$

- **Practically analysis in RD & milk Unit**

The strawberry flavored milk samples were analyzed for total solids contents. For this purpose, strawberry flavored milk sample was thoroughly mixed and 2-3g was transferred to solid analyzer pre-weighed of sample and after test complete weight of sample.

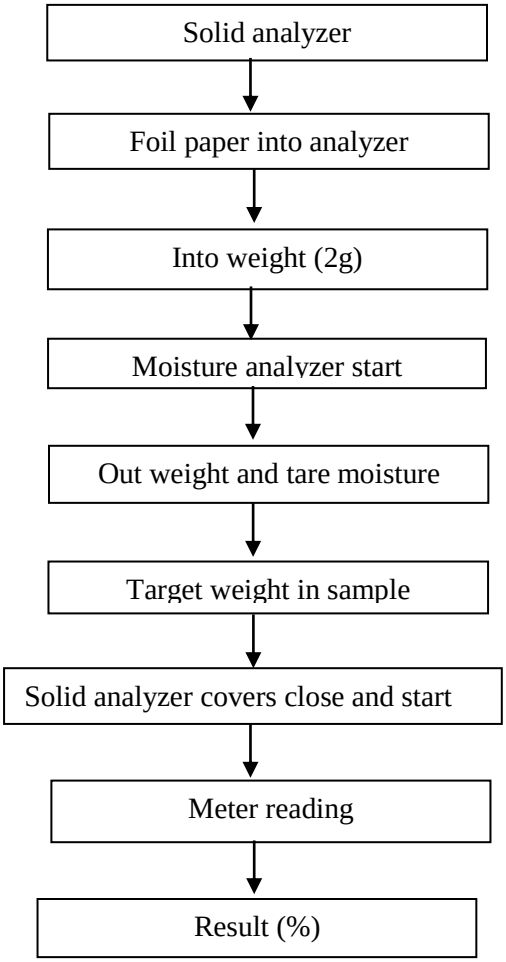


Figure: 2. Solid Analysis

3.7 Fat Analysis

- **Relative information**

Fat is the most important constituent of milk as it is used as a basis for fixing the purchase and sale price of milk. It helps to detect adulteration like water adding and skimming of milk.

In this test H_2SO_4 is used to increase specific gravity of milk serum which makes greater difference between milk serum and fat globules. It also destroys stickiness of milk by dissolving all the SNF.

The free fat globules rise to the surface by subsequent application of centrifugal force to this mixture and heat produced due to mixing of acid and milk, causing melting of fat. It facilitates the fat particles to come to the surface freely.

The specific gravity of fat is 0.9 and that of acid milk mixtures is 1.43. This situation promotes complete separation of fat when proper centrifugal force is applied. Due to application of centrifugal force lighter substances (Butter fat) are thrown towards centre and rest of serum portion that is heavier is thrown towards the pheriphen. Addition of amyl alcohol helps for separation of fat from the milk acid mixture and also prevents the charging of fat and sugar by the H2SO4.

- **Gerber method of fat analysis**

Fat content was determined by Gerber method (James, 1995). Milk sample (10.75 ml) was mixed with commercial sulfuric acid (10 ml) and amyl alcohol (1ml) in butyrometer and closed with rubber cork. Sample was centrifuged in Gerber centrifuge machine for 5 mins at 1100 rpm. The fat percentage was noted on the butyrometer scale. Solids not fat (SNF) content was determined by difference as reported by Harding (1995), using the following formula:

SNF content (%) = TS (%) – Fat (%)

- **Practically fat analysis in RD & Milk Unit**

The chocolate flavored milk samples were analyzed for fat contents. For this purpose of total fat analysis.

At first H_2SO_4 content 10 ml into butyrometer, than add 10.75 ml milk sample with content of 1 ml amyl alcohol. Then shaking are properly .When mixed complete than insert into centrifugal machine and for 5 min.

Butromatae

Test procedure

- Reagent

H_2SO_4 10ml (95%), Amyl Alcohol 1 ml (100%).

- Apparatus Required

10.75 ml pipette, 8% Butyrometer, Centrifuge machine

Centrifuge

- Procedure

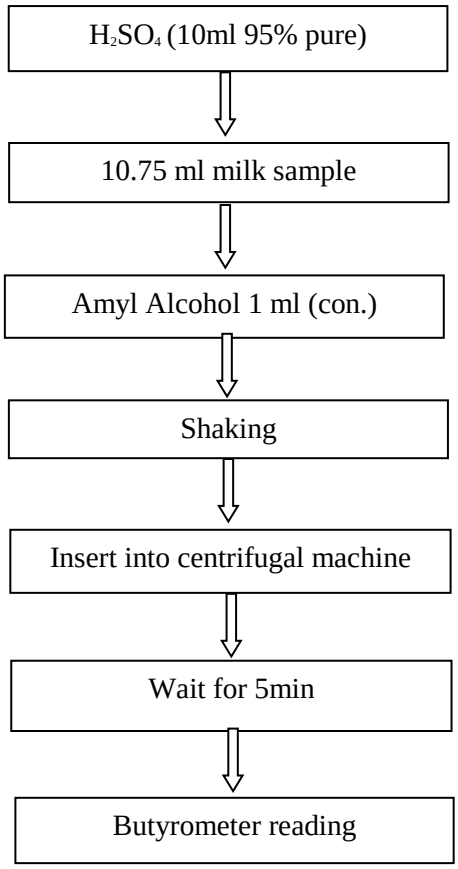


Figure: 3.Fat Analysis

3.8 Protein Analysis (3.2g)

Protein content was determined according to the method of British Standards Institution (BSI, 1990). The sample (5-6g) was digested using micro-Kjeldhal digester in the presence of catalyst (0.2g copper sulfate and 2g sodium sulfate/potassi um sulfate) where sulfuric acid (20-30 ml) was used as an oxidizing agent. The digested sample was diluted with distilled water (250 ml). Then 5 ml portion from the diluted sample was distilled with NaOH (40%), using micro-Kjeldhal distillation unit where steam was distilled over 2% boric acid (5 ml) containing an indicator for 3 minutes. The ammonia trapped in boric acid was determined by titrating with 0.1N HCl. The nitrogen percentage was calculated using following formula:

Nitrogen (%) = 1.4 (HCL, sample - HCL, blank) Normality

HCL/Weight of sample taken

The protein percentage was determined by conversion of nitrogen percentage to protein assuming that all the nitrogen in milk was present as protein i.e.

Protein percentage = N (%) × F

Where, conversion factor "F" = 100/N (N %) in dairy products (i.e. 15.66) (James, 1995).

Casein content was determined according to the method of AOAC (1990). Ash (%age) was determined by Gravimetric method, as described by AOAC (1990) using muffle furnace at 550oC. The lactose content was determined according to the following formula:

Lactose (%) = TS% - (Fat% + Protein% + Ash %)

Agencies, since water is lighter than milk and its addition would lower the specific gravity of milk. The DF milk had the lowest freezing point, varying between -0.52 and -0.57, while the other sources had lower values of freezing point of milk. These results are in agreement with the work of Shah (1996) and Prasad (1997). A number of factors including individuality, breed differences, developed acidity, colostrums, mastitis, and stage of lactation, nutrition and season can affect freezing point of milk (Packard, 1995).

3.9 Acidity Test

Acidity percentage was determined according to the method of Association of Official Analytical Chemists (AOAC, 1990). The samples were titrated with N/10 NaOH solution using titration kit with phenolphthalein as an indicator. The volume of alkali used was noted, and calculation was made using following formula:

$$\text{Titrable acidity (\%)} = \frac{\frac{N}{10}\,\text{NaOH (ml)} \times 0.009}{\text{Weight of milk sample}} x100$$

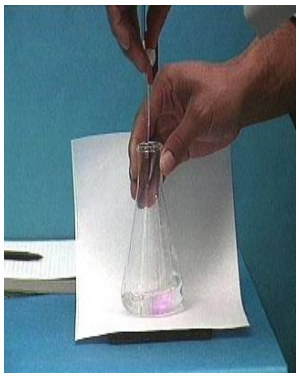
- Apparatus
- Beaker, burette, pipette
- Reagent

NaOH (0.1 N)

- Indicator

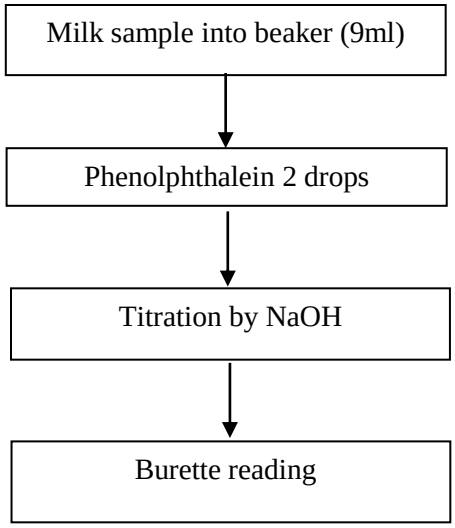
Phenolphthalein

Burette



Titration

- Test procedure



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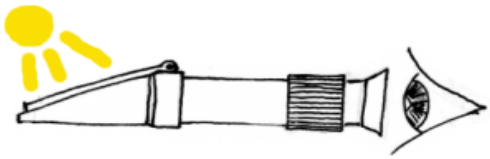
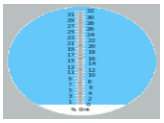


3.10 Determination °Brix



Apparatus

Refractor meter

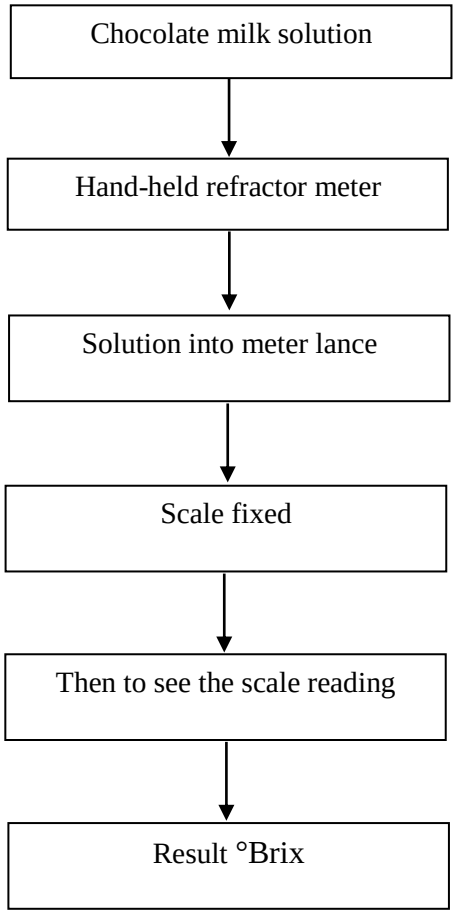


Figure: 5. ° Brix Analysis

3.11 Chemical quality

Total solids content of DF milk in the present study averaged 16.30 ± 0.14%. Similar value was reported by Prasad (1997). However, Memon (2000), Chaudhry (2002) and Inayat (2002) reported lower total solids contents (between 14.60 and 15.78%). Total solids contents of milk

collected from different channels averaged between 13.45 and 14.18, and were significantly (P<0.001) lower than control group.

Solid not fat (SNF) contents of milk obtained from DF averaged 9.79 ± 0.10% (Table 2), and did not meet the reported work of Prasad (1997) and Inayat (2002). Average SNF content of DF milk is relatively within the prescribed standard of Pakistan Pure Food Rule 1965 (Awan, 2000). SNF contents of milk obtained from different agencies were significantly (P<0.001) lower than control. These results do not meet the legal minimum standard of Pakistan Pure Food Rule for buffalo milk (9.00%) but relatively similar to that of cow milk (8.50%). It appears that different agencies at Tandojam sale either pure cow milk or it was adulterated with buffalo milk. However, all the physical attributes of milk from these agencies particularly freezing point recorded in the e present investigation also suggest water adulteration.

Fat content of milk obtained from DF varied between 5.90 and 7.20%, with an average of 6.51 ± 0.06%. These results are in line with earlier studies (Prasad, 1997; Memon, 2000; Chaudhry, 2002; Inayat, 2002). However, the fat contents of milk obtained from various channels were relatively similar (P>0.05) and were significantly lower (P<0.01) from control group. Variation in the fat content of milk might be due to individuality of buffaloes caused by differences of capacity under the identical conditions of environment, feeding and management. Such differences of individuality are better thought of as the result of genetic factors (Harding, 1995; Prasad, 1997).

Protein contents of milk obtained from DS, MCC and MVS were relatively similar and in line with the values reported by Prasad (1997). However, these were lower compared to control group.

The higher percentage of protein content in DF milk could be attributed to manage mental practices. However, hotel milk was relatively similar to that of dairy farm milk (P>0.05) in protein contents.

Casein content of milk obtained from DS, MCC, and MVS averaged 2.70, 2.77 and 2.56%, respectively. These findings were significantly lower than DF (P<0.01). However, casein content of hotel milk was relatively similar (P>0.05) to that of dairy farm milk. Milk available at hotels

is usually heated which May denaturize whey proteins (Packard, 1995) and results in the higher amount of casein.

Lactose content was highest in DF milk and lowest in milk from vendor shop. Similar findings were reported by Prasad (1997). However, the lactose content of milk obtained from DS, MCC, MVS and HT were significantly different from control group ($P<0.01$). Sharif, et al. (2007) recorded a reduction of the lactose contents in buffalo milk with the severity of sub-clinical mastitis.

Ash content of DF milk averaged 0.91% in contrast to DS (0.75%), MCC (0.78%), MVS (0.74%) and HT (0.76%) milk. The ash contents of milk collected from DS, MCC, MVS and HT were significantly lower ($P<0.01$) compared to DF milk.

3.12 Physical analysis

- p^H Analysis

The p^H of milk was recorded using a p^H meter (Hanna Instruments, HI 8417, and Italy). The instrument was first calibrated using buffers of p^H 7.0 and 4.0. Then the p^H of samples was measured.

- **Apparatus**

p^H Meter, beaker

- **Procedures**

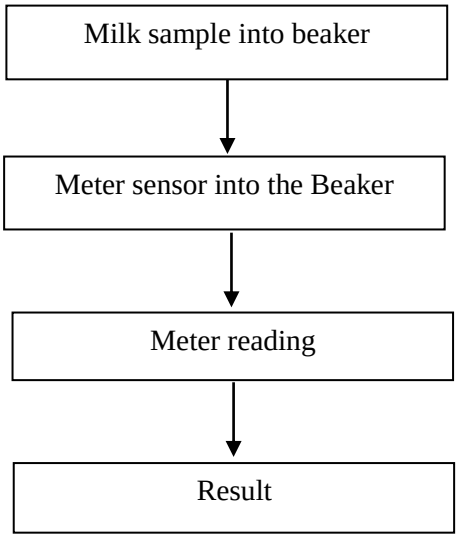


Figure: 6. pH Analysis

- **Color**

Chocolate flavored milk is a liquid of chocolate color. Color varies from bluish coffee to light chocolate, depending upon the breed of the cow, the feed fed to the cow, and the quantity of fat and other solids present in it Cow's milk is yellow white that of buffalo sheep, goat and other Spp. is white. Yellow color of the milk is due to a pigment known as carotene which is synthesized from the green feed fed to the cow

Conversion of carotenes into Vita chieflyoccures in liver. In case of buffaloed milk this change is complete and thus buffalo's milk is white.

In case of cows this conversion of carotene into Vit. A is partial so cow's milk is yellow in color. The white color (adolescence) of milk is duet reflection of light by the fat globules and the colloidal protein, calcium casein ate & phosphate

- **Taste**

Chocolate flavored milk is sweet and chocolate flavored in taste. This is due to the presence of lactose (Milk Sugar) in it with sugar. The sweet lactose is balances against the salty taste of chloride in milk.

- **Smell**

Chocolate flavored milk has got a characteristic chocolate flavored.

- **Specific Gravity**

Specific gravity of milk was determined by using pycnometer (AOAC, 1990). The density of milk was measured against the density of standard (water). Firstly, reweighed pycnometer was filled with standard reference fluid (water) to some predetermined level at 20° C and weight was noted. Then, milk sample was filled in a similar pycnometer to similar level and temperature, and weighed. Specific gravity of milk was calculated by the following formula:

$$\text{Specific gravity} = \frac{\text{Sample weight}}{\text{Total weight}} \times 100$$

- **Viscosity**

Viscosity of milk was determined with viscometer at 20oC (AOAC, 1990). The rate of flow of a given volume of milk was compared with the rate of flow of the same amount of water. The viscosity of milk was calculated according to the following formula:

Viscosity (cP) = Flow time of milk at 20oC × Specific gravity of milk ×1.002/flow time of water at 20°C. Freezing point of milk was determined using cryoscopy (AOAC, 1990).

- **Physical quality of milk**

There was significant difference (P<0.01) in the acidity percentage of DF mi elk compared to the other sources (Table 1). However, the acidity of milk was relatively similar (P>0.05) at DS vs. MCC, MVS and HT, MCC versus MVS and HT. Wherever, milk obtained from MVS was significantly different (P<0.05) from that of HT milk. Shah (1996) also reported similar trend for acidity of milk sold by different agencies at Latifabad, Hyderabad.

Mean pH of milk obtained from various sources varied between 6.53 and 6.66 and was within the normal range (Table 1). Relatively similar observations were made by different workers (Memon, 2000; Inayat, 2002). The results of terrible acidity and pH values of milk in the present study are not correlating to each other. The reason could be attributed to addition of water, ice or chemical preservative in pure raw milk to extend its shelf life.

Viscosity of DF milk in the present study varied between 1.61 and 2.10 centipoises (cP) with an overall average of 1.86 ± 0.02 cP (Table 1). These results are in line with the study conducted by Prasad (1997). However, the milk obtained from DS, MCC, MVS and HT showed lower viscosity (Table 1). Statistically, viscosity of milk from various channels was signify- cantle lower (P<0.001) than control group. This can be attributed to condition and quality of protein or fat (i.e. percentage, size of fat globules or clusters of fat globules in milk), temperature e and period of storage of milk and/or water adulteration (Prasad, 1997).

The DF milk showed the highest specific gravity (1.031 ± 0.001), while the specific gravity of milk obtained from other agencies was significantly (P<0.001) lower (~1.026) than control group. These results suggest the water adulteration in milk from these.

- **Sensory evaluation**

The consumer's acceptability of development products was evaluated by a taste testing panel. The hedonic rating was used to determine this acceptability .The score was used for judging the products. The panelists were asked to rate their acceptability for the product on the hedonic scal.1 – 9, ranging from "dislike extremely to like extremely. Some numerical value

Food companies regularly use sensory tests, such as descriptive analysis and consumer affective tests, to study ingredient effects, processing variables and storage changes on the perceived

sensory properties of their products. Sensory analysis provides marketers with an understanding of product quality, directions for product quality, profiles of competing products, and evaluations of product reformulations from a consumer perspective (Stone and Sidel, 1993).

Sensory evaluation techniques include: 1) Descriptive Sensory Analysis, 2) Discriminative Sensory Analysis, and 3) Consumer affective tests.

Descriptive Sensory Analysis has been widely used to characterize in detail aroma, flavor, and oral texture attributes of food products. All descriptive analysis methods involve the objective detection, description and quantification of sensory attributes of a product by trained panelists (Meilgaard and others, 1999). Descriptive sensory techniques include Quantitative Descriptive Analysis (QDA®), Flavor Profile Analysis (FPA), Texture Profile Analysis (TPA), Free Choice Profiling, and Spectrum TM Descriptive Analysis.

Among the discriminating sensory tests, paired comparison, duo-trio test, and triangle test are the most commonly used? Basically they are utilized to detect differences between two or more food products (Stone and Sidel, 1993).

Consumer affective tests can help the sensory scientist to understand the behavior of different consumer's groups (Piggott, 1988), and therefore to understand potential buyers of the product and in which way such a product can be inserted into the food market. Data obtained.

From consumer affective tests represent key information in studies of product development, quality control, food product acceptance, and food service evaluation (Piggot, 1988). There are two types of affective tests, quantitative and qualitative. Qualitative tests (i.e., focus group interviews, focus panel, one-on-one interviews) measure subjective responses of a small group of representative consumers to the sensory properties of products by having them talk about their feelings in an interview or group setting (M eilgaard and others, 1999). Quantitative tests determine the responses of a large group of consumers to a set of questions regarding preference, liking, sensory attributes, etc. (Meilgaard and others, 1999).

The consumer's acceptability of developed products was evaluated by a taste testing panel. The hedonic rating test was used to determine this acceptability. The score was used for judging the products. The score was shown in Appendix II. The panelists were selected from

different social status. The panelists were asked to rate their acceptability for the product on a hedonic scale, 1 to 9, ranging from “dislike extremely” to “like extremely”. Some numerical values are assigned to each point on the scale to analyze the result using statistical methods. The scale was arranged such that: 9 = Like extremely; 8 = Like very much; 7 = Like moderately; 6 = Like slightly; 5 = Neither like or dislike; 4 = Dislike slightly; 3 = Dislike moderately; 2 = Dislike very much and 1 = Dislike extremely.

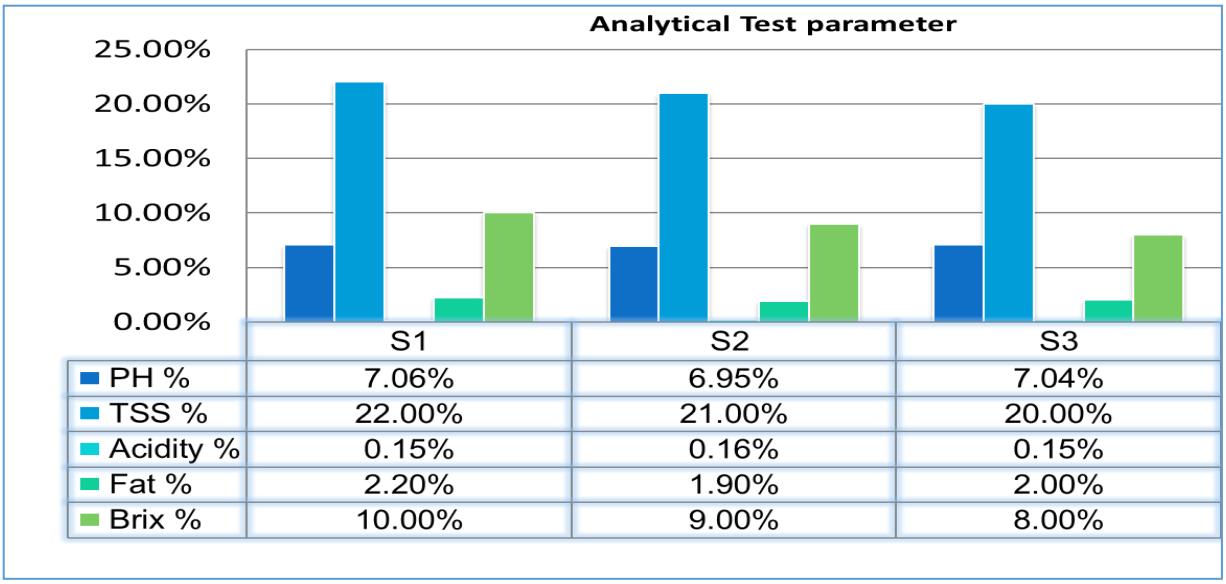
3.13 Shelf life analysis

Chocolate flavored milk was collected in Pouches pack and stored at 4 °C for 10 days for shelf-life studies. Each bag sample was used and discarded after analysis. Every other day, the physicochemical characteristics (pH and color) also flavored were evaluated. The microbial loads were poured into Plate Count Agar to count the mesophilic aerobes. Samples were taken from a different bag each day, diluted with peptone water and poured into dishes. Samples were incubated at 35 °C for 48 h. Every three days the coloring, flavored, chemical and microbiology agent done evaluation.

CHAPTER IV

RESULT & DISCUSSION

4.1 RESULT & DISCUSSION



	S1	S2	S3
PH %	7.06%	6.95%	7.04%
TSS %	22.00%	21.00%	20.00%
Acidity %	0.15%	0.16%	0.15%
Fat %	2.20%	1.90%	2.00%
Brix %	10.00%	9.00%	8.00%

➢**Fat:** All samples were used pasteurized milk. We were finding the same fat in chocolate flavored milk samples. Pasteurized milk content 3.5% Fat.

➢**°Brix:** Each sample we have done °Brix determination. We were finding the result of S1, S2, and S3 sample first sample sugar used 8.68%, second sample 7.68% and third sample 6.68% extra sweetness content from chocolate fruit concentrate.

➢**Acidity:** Acidity depended milk lactose and chocolate powder concentrate. All samples a same quantity of milk & chocolate powder concentrate .There for finding the result were same.

➢**Total soluble solid:** Chocolate flavored milk Solid mainly depended on sugar % and chocolate powder concentrate .Solid depend mainly sugar. There for finding the result were same varying.

➢**p^H:** p^H mainly deepened on alkalinity and acidy Milk content lactic acid .All sample were used are same milk. So we found are same pH range.

4.2 Packaging Study

Table: 3. organoleptic assessment of flavored milk during storage at freezing temperature in different packaging system.

Storage condition	Storage period (days)	Samples	Colour	Flavour	Texture
Pouches pack	0	S_1	Light chocolate	Pleasant	Slight thick
		S_2	Light chocolate	Pleasant	Slightly less thick
		S_3	Light chocolate	Pleasant	Slightly less thick
	10	S_1	Light chocolate	Pleasant	Slight thick
		S_2	Light chocolate	Less pleasant	Slightly less thick
		S_3	Light chocolate	Less Pleasant	Slightly less thick
Tetra pack	0	S_1	Light chocolate	Pleasant	Slight thick
		S_2	Light chocolate	Pleasant	Slightly less thick
		S_3	Light chocolate	Pleasant	Slightly less thick
	10	S_1	Light chocolate	Pleasant	Slight thick
		S_2	Light chocolate	Pleasant	Slightly less thick
		S_3	Light chocolate	Less pleasant	Slightly less thick

4.3 Sensory evaluation

The samples of chocolate flavored milk were subjected to sensory evaluation.

Thus 3 samples with different sugar ratio were evaluated by panelists for color, flavor, texture and overall acceptability using 1-9-point hedonic scale. The mean scores for these quality attributes together with analysis of variance are given in Table 4.

Table: 4. Mean scores for color, flavor, texture and overall acceptability of flavored milk.

Sample	Sensory attributes			
	Color	Flavor	Texture	Overall acceptability
S_1	8.18^A	8.09^A	8.18^A	8.00^A
S_2	7.72^B	7.18^B	7.45^B	7.45^B
S_3	7.45^B	7.45^B	7.36^B	7.36^B

Sample S_1 = Milk= 2385 g; Sugar = 267.95g; Milk gel=6.36g; SMP=75g; Chocolates powder concentrate = 159g; Chocolate color = 0.04g; Chocolate flavor= 1.2g; Water= 90g.

Sample S_2 = Milk= 2385 g; Sugar = 237.08g; Milk gel=6.36g; SMP=75g; Chocolate powder concentrate = 159g; Chocolate color = 0.04g; Chocolate flavor= 1.2g; Water= 90g.

Sample S_3 = Milk= 2385 g; Sugar = 212.55g; Milk gel=6.36g; SMP=75g; Chocolate powder concentrate = 159g; Chocolate color = 0.04g; Chocolate flavor= 1.2g; Water= 90g.

A two way analysis of variance (ANOVA) was carried out for colour preference flavored milk and the result showed (Table 4) that sample S_2 and S_3 were equally acceptable and differed significantly from the sample S_1 and S_1 was the most preferred one. Sample S_1 secured the highest score 8.18 out of 9 points for colour and can be ranked as “like very much”. Samples S_2 and S_3 are comparatively less acceptable securing scores 7.72 and 7.45 respectively and both can be ranked as “like moderately”.

In case of flavor the results showed that there were significant differences in flavor acceptance of the sample S_2 and S_3 with the sample S_1 as the calculated F value (18.8095) was greater than the tabulated F value (3.49).

As shown in Table 4 sample S_1 was the most acceptable in flavor preference among the samples securing the highest score (8.09) and ranked as “like very much”. This was followed by sample S_2 and S_3 securing 7.18 and 7.45 respectively and were equally acceptable. Thus, samples S_2, and S_3 can be ranked as “like moderately”.

In case of texture of products there were no significant differences between the samples S_2 and S_3 but sample S_1 was different securing highest score (8.18) and keeping ranked as “like very much”. Sample S_2 and S_3 securing score 7.45 and 7.36 respectively and keeping ranked as “like moderately”.

Results (Table 4) revealed that there was no significant difference for overall acceptability between samples S_2 and S_3 securing 7.45 and 7.36 respectively and these samples can be ranked as “like moderately”. It is also seen that sample S_1 was the most acceptable product securing the highest score 8.0 out of 9 among the samples and ranked as “like very much”.

From the above results, it is clearly seen that with 267.95g sugar treated sample S_1 was the most preferred milk with respect to all the quality attributes and as such ranked as “like very much”. Though the other samples could be ranked as “like moderately”, sample S_2 and sample S_3 secured the lowest score indicating that were least preferred compared to sample S_1 and between the samples S_2 and S_3, there were no significant difference.

CHAPTER V

CONCLUSION

Schools should keep serving chocolate flavored milk. There should be chocolate flavored milk because kids like it, it gives vitamins, and it gets kids in good habits. Many kids love chocolate flavored milk – it makes them happy to see it in the cafeteria, their lunch box, at their kitchen table. Research shows that, overall, chocolate flavored milk is pretty good for kids. It's especially important that kids like chocolate flavored milk. It turns out that more kids drink milk, when they can get chocolate flavored milk.

When you interview a lot of parents, like Katie Couric did, they'll say that their kids only drink milk if they can get chocolate flavored milk. So at least they're drinking milk. In a survey of students in this school, 84% said that they would drink more milk if they had chocolate flavored milk available. Of those same students, 28% said that they wouldn't drink any milk at all unless it was chocolate flavored.

Surprisingly, chocolate flavored milk turns out to have vitamins. A nutritionist from the Dairy

Association, demonstrates that chocolate flavored milk is a good source of vitamin A, D, E, and calcium.

That's a lot of vitamins and they're in something kids actually like to drink! In her information

Session, the nutritionist is with kids who drink chocolate flavored milk. Their bright teeth and glossy hair

Illustrates that kids who love chocolate flavored milk will be that healthy.

CHAPTER VI

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APPENDIX

Table 1.1 Sensory evaluation data for flavored milk

Judge no.	S_1	S_2	S_3	Total
1	8	8	7	23
2	9	8	8	25
3	7	7	7	21
4	8	7	6	21
5	8	8	8	24
6	9	8	8	25
7	8	7	8	23
8	9	8	7	24
9	8	8	8	24
10	8	8	7	23
11	8	8	8	24
Total	90	85	82	257
Avg.	8.182	7.727	7.455	

Table 1.1.2 ANOVA (Analysis of variance) for color

Sources of variance	Degree of freedom	Sum of squares	Mean squares	F-value	
				Calculated	Tabulated (5%)
Products	2	2.970	1.485	6.8056	3.49
Judges	10	6.182	0.618	2.8333	2.365
Error	20	4.364	0.218		
Total	32	13.515			

Table 1.1.3 Duncun's Multiple Range Test (DMRT) for color

LSD = 0.4153; $P<0.05$

Sample	Original order of means	Sample	Ranking order of means
S_1	8.182	S_1	8.182
S_2	7.727	S_2	7.727
S_3	7.455	S_3	7.455

Table 1.1.4 Rating score for flavor of flavored milk

Judge no.	S_1	S_2	S_3	Total
1	8	7	7	22
2	8	7	7	22
3	7	7	7	21
4	8	7	8	23
5	8	7	7	22
6	8	7	7	22
7	9	8	8	25
8	8	8	8	24
9	9	7	8	24
10	8	7	7	22
11	8	7	8	23
Total	89	79	82	250
Avg.	8.091	7.182	7.455	

Table 1.1.5 ANOVA (Analysis of variance) for flavor

Sources of variance	Degree of freedom	Sum of squares	Mean squares	F-value Calculated
Products	2	4.788	2.394	18.8095
Judges	10	4.727	0.473	3.7143
Error	20	2.545	0.127	
Total	32	12.061		

Table 1.1.6 Duncun's Multiple Range Test (DMRT) for flavour

LSD =0.3170; $P<0.05$

Sample	Original order of means	Sample	Ranking order of means
S_1	8.091	S_1	8.091
S_2	7.182	S_3	7.455
S_3	7.455	S_2	7.182

Table 1.1.7 Rating score for texture of flavored milk

Judge no.	S_1	S_2	S_3	Total
1	8	7	7	22
2	8	8	7	23
3	9	8	7	24
4	8	7	7	22
5	8	7	7	22
6	8	8	8	24
7	8	8	8	24
8	8	8	7	23
9	8	7	8	23
10	8	7	7	22
11	9	7	8	24
Total	90	82	81	253
Avg.	8.182	7.455	7.364	

Table 1.1.8 ANOVA (Analysis of variance) for texture

Sources of variance	Degree of freedom	Sum of squares	Mean squares	F-value	
				Calculated	Tabulated (5%)
Products	2	4.424	2.212	10.4286	3.49
Judges	10	2.667	0.267	1.2571	2.365
Error	20	4.242	0.212		
Total	32	11.333			

Table 1.1.9 Duncun's Multiple Range Test (DMRT) for texture

LSD = 0.4095; $P<0.05$

Sample	Original order of means	Sample	Ranking order of means
S_1	8.182	S_1	8.182
S_2	7.455	S_2	7.455
S_3	7.364	S_3	7.364

Table 1.1.10 Rating score for overall acceptability of flavored milk

Judge no.	S_1	S_2	S_3	Total
1	8	8	7	23
2	8	7	7	22
3	8	8	7	23
4	8	7	7	22
5	8	8	8	24
6	8	8	8	24
7	8	8	8	24
8	8	7	7	22
9	8	7	8	23
10	8	7	7	22
11	8	7	7	22
Total	88	82	81	251
Avg.	8.000	7.455	7.364	

Table 1.1.11 ANOVA (Analysis of variance) for overall acceptability

Sources of variance	Degree of freedom	Sum of squares	Mean squares	F-value	
				Calculated	Tabulated (5%)
Products	2	2.606	1.303	9.5556	3.49
Judges	10	2.545	0.255	1.8667	2.365
Error	20	2.727	0.136		
Total	32	7.879			

Table 1.1.12 Duncun's Multiple Range Test (DMRT) for overall acceptability

LSD = 0.3280; $P<0.05$

Sample	Original order of means	Sample	Ranking order of means
S_1	8.000	S_1	8.000
S_2	7.455	S_2	7.455
S_3	7.364	S_3	7.364

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Signature